Yuhao Deng

AI Engineer & Architect | AI Agents, Memory Systems, Rustacean

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China
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Summary

AI engineer and architect with 5+ years of software engineering experience across Ericsson, Alibaba Cloud, and Super Intelligence Inc. Specializes in AI agent runtimes, agent memory building, and reliable backend systems. Led Megahit's backend and agent architecture design, including durable task execution and behavioral evaluation. Builds in Python, Rust, and C++, balancing performance, delivery speed, and reliability with product needs.

Skills

C++

STL, C++17, Linux, CUDA

Rust

Tokio, Axum, SQLx, MCP

Python

FastAPI, PydanticAI, PyTorch, pandas, AI agents

TypeScript

React, AI agents

Infrastructure

AWS, Alibaba Cloud, PostgreSQL, Kubernetes, CI/CD

Certifications

Alibaba Cloud Certified Professional - Cloud Computing
Dec 2025
Alibaba Cloud

Alibaba Cloud Certified Professional - LLM
Oct 2025
Alibaba Cloud

Languages

Chinese
Native

English
B2

French
B1

Interests

Philosophy

Historical European Martial Arts (HEMA)

Linguistics

Education

University of Electronic Science and Technology of China Master's Degree
Signal and Information Processing 2018 – 2021

Hebei University of Technology Bachelor's Degree
Electronic Information Engineering 2014 – 2018

Experience

Super Intelligence Inc. Jul 2026 – Nov 2026
AI Engineer & Architect

- Owned backend architecture for Megahit, an AI growth platform; led the staged migration from a custom agent loop to PydanticAI and retired the legacy runtime.
- Migrated Supabase to Aurora PostgreSQL and maintained separate API and worker services on AWS ECS. Automated build/deployment checks and refactored backend CI/CD.
- Built a Rust/Axum authorization service with Cognito OIDC and PostgreSQL, covering session revocation, tenant membership, explicit action grants, and transactional audit records; completed integration and acceptance in the AWS.
- Built a Rust MCP data workspace with 17 structured-data and file tools, per-owner database isolation, scoped access, idempotent mutations, bounded queries, and authenticated file transfers.

Alibaba Cloud Hangzhou
Site Reliability Engineer (SRE) Sep 2025 – Jun 2026

- Handled on-call incidents for cloud services and distributed databases: triaged alerts, assessed customer impact, and coordinated escalation and mitigation with engineering teams.
- Performed health checks, operational-readiness reviews, and Kubernetes troubleshooting. Developed emergency-response and drill-platform tooling in Rust and TypeScript, with CI/CD and operational automation.
- Worked with international customers, support teams, and engineers to investigate technical issues, communicate incident updates, and coordinate follow-up actions.

Ericsson Chengdu
Experienced Developer Software Jul 2021 – Sep 2025

- Developed Linux-based radio software in C++17, focusing on signal-processing libraries, numerical algorithms, and performance-sensitive systems.
- Implemented concurrent and asynchronous components, optimized matrix computations, and investigated floating-point precision and runtime issues through low-level debugging.

Projects

Megahit — AI Agent Architecture & Evaluation Jul 2026 – Nov 2026

- Designed service boundaries and frontend/backend contracts for conversational agents and campaign planning/execution across FastAPI, React, AWS ECS, and Aurora PostgreSQL. Separated model reasoning from deterministic business commands, with PostgreSQL as the business source of truth.
- Decoupled long-running tasks from browser/SSE connections. Added durable checkpoints, worker leases/fencing, and a persistent activity feed for restart recovery; used content-bound approvals and an idempotent operation ledger to guard scheduled publishing against duplicate actions.
- Partnered with a memory researcher to integrate shared memory into growth agent, preserving provenance while isolating execution state and authorization scopes.
- Built routing regression tests and multi-turn cases for form submission, memory correction, draft revision, and concurrent requests. Ran integration tests in CI and real-model behavioral evaluations; one completed run covered 773 queries with code/model metadata recorded. Fixed approval expiry to cover the full publishing schedule.

Product (login required)

cc-switch-cli — Open-Source Contributor Apr 2026 – Jun 2026

- Contributed 12 merged PRs to a Rust CLI/TUI for AI coding assistant provider and proxy management, covering daemon supervision, failover, configuration safety, and protocol compatibility.
- Implemented a Rust daemon with IPC and supervised proxy workers, keeping proxies running after CLI/TUI exit and supporting restart recovery (PR #189).
- Added prioritized provider failover with TUI controls, scriptable CLI commands, and mock-provider integration tests. Fixed configuration merging and hot-switch backups to preserve local edits and restore the latest selected provider.
- Improved OpenAI/Anthropic streaming and tool-call compatibility; isolated filesystem/process state in parallel tests and added unit and integration suites to CI.

https://github.com/SaladDay/cc-switch-cli

Time-Series Forecasting — Model Evaluation & Deployment Apr 2025 – Sep 2025

- Led model comparison, training, hyperparameter tuning, and deployment for an internal forecasting task; evaluated LSTM, LightGBM, and Informer.
- Selected Informer for long-range temporal dependencies and sparse attention. Achieved RMSE of approximately 3 on the internal dataset; the project won second prize in the company AI Hackathon.

C++ Signal-Processing Library — Architecture & Optimization Jun 2024 – Mar 2025

- Redesigned inter-thread communication using Protocol Buffers and Linux epoll, converting key components from synchronous calls to asynchronous communication.
- Parallelized matrix factorization and rewrote Cholesky decomposition, delivering approximately 6× faster matrix operations in key workloads.
- Used objdump and disassembly analysis to diagnose floating-point precision and runtime issues.